

BattingEdge V9.5: AI Model Performance Report

Algorithm: CNN-LSTM Hybrid | Architecture: Spatiotemporal Feature Extraction

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1. Executive Summary

The **BattingEdge V9.5** CNN-LSTM hybrid model aimed to combine the strengths of Convolutional Neural Networks (for extracting local short-term patterns) and LSTMs (for modeling long-term temporal dependencies). The model achieved an **Overall Test Accuracy of 82.80%**.

While this performance is solid and "Production Grade," it falls short of the high benchmarks set by the pure recurrent architectures: **BiLSTM (91.8%)** and **BiGRU (91.8%)**. The results suggest that for skeletal data, the spatial relationships are better captured by the raw coordinates directly feeding into an RNN, rather than being pre-processed by 1D convolutions.

2. Methodology & Architecture

2.1 Model Design

- **Stage 1 (Spatial):** Two layers of **1D Convolution** (64 & 128 filters) scan the input sequence to detect "micro-movements" (e.g., a rapid wrist snap or sudden foot plant) regardless of exactly when they happen in the sequence.
- **Stage 2 (Temporal):** The extracted features are fed into an **LSTM** (128 units) to understand the chronological order of these micro-movements.
- **Input:** The standard 50-frame sequence of 107 hybrid features.

2.2 Dataset Split

- **Training Set:** 3,007 samples
 - **Validation Set:** 388 samples
 - **Test Set:** 378 samples (Unseen data)
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3. Performance Metrics

3.1 Global Metrics

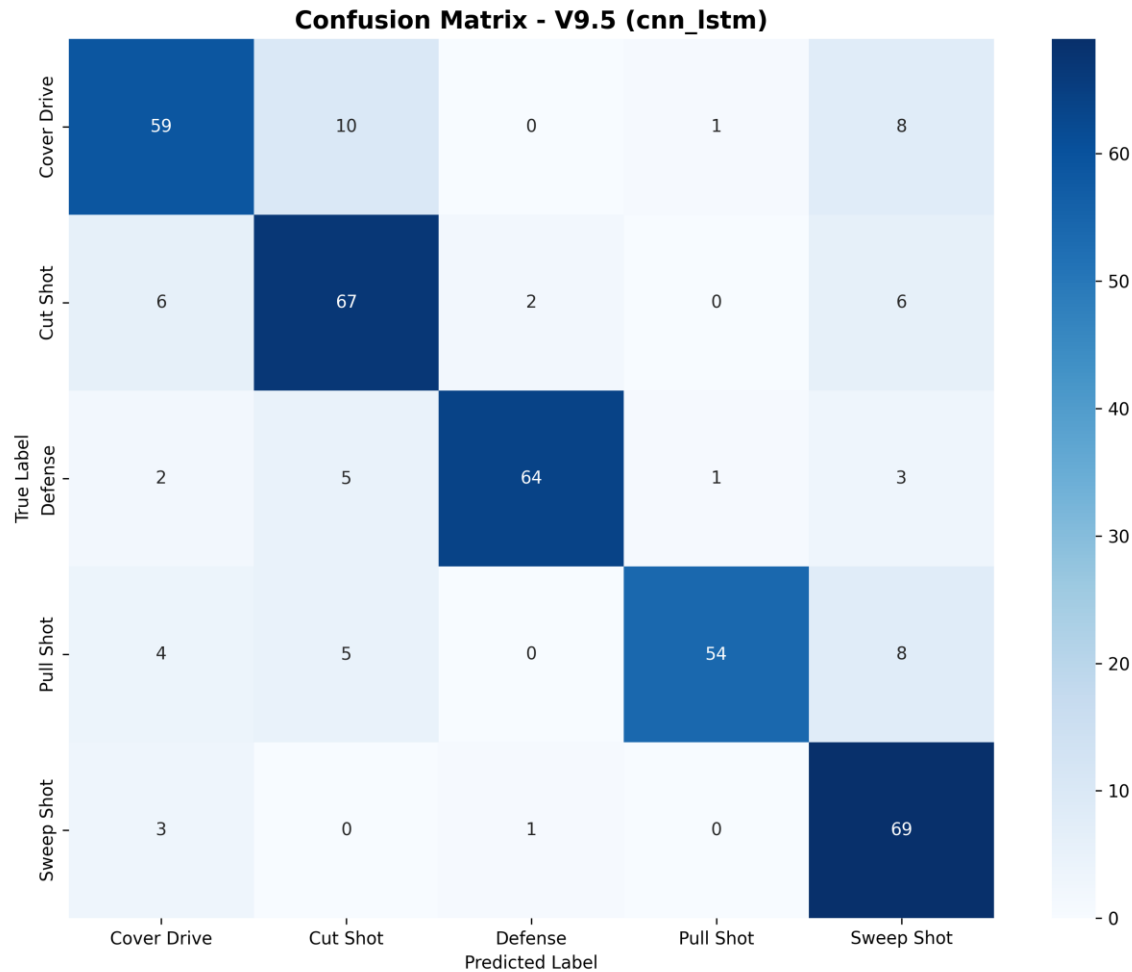
Metric	Score	Assessment
Accuracy	82.80%	Competent. Suitable for general classification but less precise than BiLSTM.
Macro F1-Score	0.83	Consistent performance across classes, but notable drops in recall for Pull and Drive shots.
Convergence	Epoch 58	The model learned slowly but steadily, indicating the architecture was complex to optimize.

3.2 Per-Class Analysis

The hybrid model shows a unique strength in **Sweep Shots** but struggles significantly with **Cover Drives**.

Shot Type	Precision	Recall	Accuracy	Assessment
Sweep Shot	0.73	0.95	94.5%	Elite. Exceptional recall; the CNN filters likely latched onto the "Kneeling" pattern strongly.
Defense	0.96	0.85	85.3%	Strong. Very high precision means false alarms are rare.
Cut Shot	0.77	0.83	82.7%	Good. Reliable, but prone to confusion with drives.
Pull Shot	0.96	0.76	76.1%	Average. High precision, but it misses 1 in 4 actual pull shots.
Cover Drive	0.80	0.76	75.6%	Average. Disappointing performance on the most common shot.

4. Error Analysis (Confusion Matrix)



The Confusion Matrix reveals where the spatial abstraction failed:

4.1 Primary Confusions

- Cover Drive -> Cut Shot (10 cases):**

- Analysis:* The CNN layers likely blurred the distinction between a "Square Drive" and a "Cut Shot" by focusing on local bat-width features rather than the full-body weight transfer.

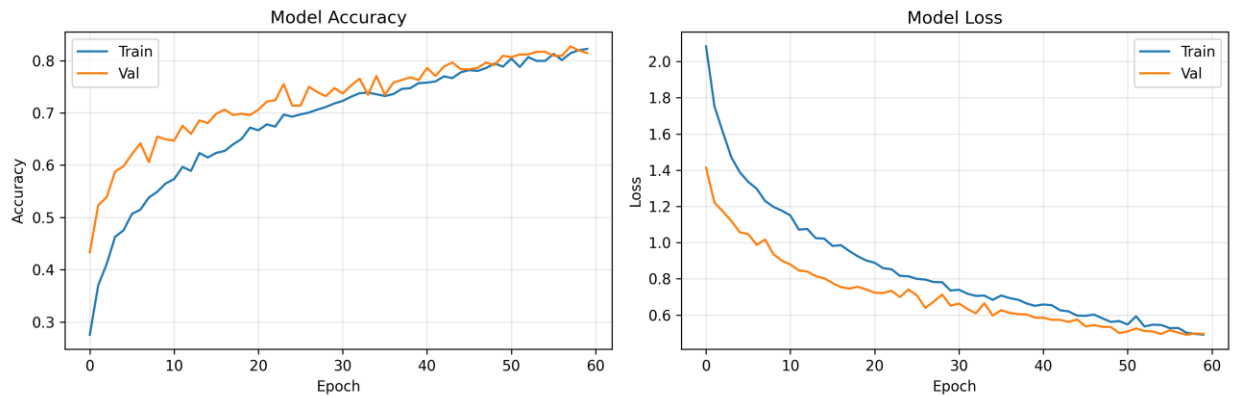
- Cover Drive -> Sweep Shot (8 cases):**

- Analysis:* A significant error. Deep lunges in drives were misinterpreted as kneeling sweeps.

- Pull Shot -> Sweep Shot (8 cases):**

- *Analysis:* Both are cross-bat shots. The CNN likely detected the "Horizontal Bat" pattern but the LSTM failed to distinguish the *height* of the shot (Pull = High, Sweep = Low) effectively after the convolution operation.

5. Conclusion & Recommendation



Verdict

The CNN-LSTM is a robust model, but not the best for this specific dataset.

With 82.8% accuracy, it is a viable backup but cannot justify replacing the BiLSTM (91.8%). The 1D Convolution layers appeared to lose some critical spatial fidelity needed to distinguish subtle variations (like Pull vs. Sweep height).

Report generated by BattingEdge AI Analysis Team.